A Multitask Benchmark Dataset for Satellite Video: Object Detection, Tracking, and Segmentation

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***Abstract*— Video satellites can continuously image large areas and provide dynamic, real-time monitoring of hotspots and objects. The intelligent processing and analysis of satellite video have become a research hotspot in the field of remote sensing. However, the lack of high-quality satellite video datasets limits the development of relevant object detection, object tracking, and object segmentation. In this article, we build the largest scale satellite video dataset with the most task types supported and object categories, named the satellite video multimission benchmark (SAT-MTB). First, multitask annotation of aircraft, ships, cars, trains, and their corresponding 14 categories of fine-grained objects in 249 satellite videos is performed based on horizontal bounding boxes (HBBs), oriented bounding boxes (OBBs), and masks, which cover more than 50 000 frames and 1 033 511 annotated object instances. Then, we review the tasks of object detection, object tracking, and object segmentation based on satellite videos, providing a comprehensive overview of progress in related datasets and algorithm research. Finally, we establish the first public benchmark of multitask algorithms for satellite video object detection, object tracking, and object segmentation, evaluating and analyzing the performance of a total of 47 representative algorithms under different tasks on the constructed dataset. The proposed SAT-MTB will significantly advance research in intelligent processing and analysis of satellite video and related applications.**

***Index Terms*— Multitask dataset, object detection, object segmentation, object tracking, satellite video.**

I. Introduction

THE development of aerospace technology has made it possible for satellites to gaze at specific areas, and the emergence of video satellites, such as Jilin-1 and the Sky-Sat series of satellites, has rapidly raised widespread interest in various fields [1], [2], [3]. Compared to conventional static remote sensing imagery, video satellite imagery with high temporal resolution has important applications in the fields of transport, security, resources, and the environment. To achieve intelligent interpretation of Earth observation, research on visual tasks of satellite video has become one of the frontiers in remote sensing [4], [5], [6], [7]. In recent years, deep learning-based artificial intelligence algorithms have made great development in basic computer vision tasks, such as object detection, object tracking, and object segmentation. Many excellent types of research have emerged, such as Faster R-CNN [8] and RetinaNet [9] for object detection, SiameseRPN [10] and KCF [11] for object tracking, and R-FCN [12] for object segmentation, which has greatly improved the capability of the image/video intelligent processing, analysis, and understanding. This is supported by large-scale open benchmark datasets, such as ImageNet [13], COCO [14], Pascal VOC [15], and OTB [16], which provide the data necessary for research, task design, and training models, as well as for the development of computer vision algorithms and the implementation of industrial applications.

Compared to conventional remote sensing imagery, satellite video has difficulties with labeling due to the small size and sparse distribution of objects, and the high cost of frame-by-frame annotations. Although some datasets, such as the single object tracking dataset SatSOT [17] and the vehicle-based multiobject detection and tracking dataset VISO [18], have been published, there still exists a significant gap in the types of tasks supported, the richness of object classes, and the number of annotations compared to image-based datasets. The lack of large-scale, unified data benchmarks for the study of different visual tasks in satellite video has become one of the main factors limiting the relevant research, which needs to be addressed urgently to promote the development of the intelligent interpretation of satellite video and breakthroughs in relevant applications.

This article constructs a large-scale, multitask satellite video dataset named the satellite video multimission benchmark (SAT-MTB), which is based on high-resolution videos acquired by the Jilin-1 satellite. We carefully select and annotate 249 video scenes with rich content, containing more than 50 000 frames. The object categories are divided into coarse-grained levels of four categories and fine-grained of 14 categories. Under the same data conditions, the horizontal

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bounding box (HBB), the oriented bounding box (OBB), and the masks of objects are annotated frame by frame, with a total of 1 033 511 annotated instances. The dataset is characterized by large-scale, multitask, fine granularity, rotating boxes, and masks. It supports the research and algorithm evaluation for various visual tasks, such as object detection, object tracking, and object segmentation of satellite video. In addition, this article provides a comprehensive review of representative methods for object detection, object tracking, and object segmentation in recent years. Furthermore, we fairly evaluate and analyze the performance of 47 representative methods for three tasks on the dataset. Finally, the characteristics and task challenges of the dataset are summarized, and an overview of potential future research directions is provided. The contributions of this article are summarized as follows.

1) This article constructs a large-scale satellite video dataset that integrates multiple tasks, such as object detection, object tracking, and object segmentation. The dataset also supports fine-grained object and OBB tasks, providing rich, high-quality data samples and benchmark evaluation conditions for the related research of satellite video.
2) A comprehensive and systematic review of datasets and methods related to various tasks, such as object detection, object tracking, and object segmentation, especially in the field of remote sensing, with analysis and perspective on future research trends.
3) This article fairly evaluates and analyzes the performance of a total of 47 advanced methods for three tasks, including object detection, object tracking, and object segmentation on SAT-MTB. We also analyze the improvement directions for each task in relation to the characteristics and challenges of the dataset.

This article is organized as follows. Section II introduces relevant dataset work in the field. Section III describes, in detail, the dataset constructed in this article and analyzes its characteristics. Section IV provides a comprehensive and systematic review of three types of task methods, including object detection, object tracking, and object segmentation, and evaluates the experimental performance of representative benchmark methods. Finally, Section V concludes and provides an outlook for the work in this thesis.

II. Related Dataset

A. Optical Remote Sensing Object Detection Dataset

1) Optical Remote Sensing Image Object Detection Dataset: Ground objects in remote sensing are characterized by the diversity of scale, specificity of viewpoint, small size, and arbitrary orientation. With the development of detection technology and the continuous improvement of spatial resolution, it is possible to refine the representation of objects, and remote sensing object detection has evolved from HBB and coarse-grained to OBB and fine-grained detection. The current remote sensing object detection dataset can be divided into HBB detection and OBB detection, for different Earth observation platforms, including satellite and airborne.

a) Object detection based on HBB: Compared with large-scale object detection datasets, such as ImageNet, MS-COCO, and Pascal VOC in the general field, the scale of object detection data of remote sensing images is much smaller due to constraints such as sparse distribution of typical ground objects and a limited number of categories, and the object detection datasets mainly take the form of HBB annotation. Representative remote sensing image object detection datasets in recent years are mainly as follows.

The DIOR [19] dataset, released by Northwestern Polytechnical University in 2020, is a large-scale benchmark dataset for optical remote sensing image object detection, containing 23 463 images and 192 472 instances covering 20 classes of objects: aircraft, airports, baseball fields, basketball courts, bridges, chimneys, dams, highway service areas, highway toll booths, ports, golf courses, ground athletic fields, flyovers, ships, stadiums, storage tanks, tennis courts, railway stations, vehicles, and windmills. The DIOR-R [20] dataset released in 2022 is an extended version of DIOR annotated with OBBs, which shares the same images with DIOR.

The HRRSD [21] dataset published by the University of Chinese Academy of Sciences in 2020 contains 21 761 images acquired from Google Earth and Baidu maps with spatial resolutions ranging from 0.15 to 1.2 m, with a total of 55 740 instances of objects, around 4000 in each category, with the 13 categories as airplanes, baseball fields, basketball courts, bridges, crossroads, athletic fields, harbors, car parks, boats, storage tanks, crossroads, tennis courts, and cars.

The LEVIR [22] dataset, released by Beihang University in 2018, consists of more than 22k images of 800 $\times$ 600 pixels from Google Earth, with spatial resolutions ranging from 0.2 to 1 m. The dataset covers most types of ground scenes, such as urban, rural, mountainous, and marine, and includes a total of three object categories: aircraft, ships, and oil tanks, with a total of 11 000 instances, including 4724 aircraft, 3025 ships, and 3279 oil tanks.

The AI-TOD [23] dataset released by Wuhan University in 2020 is a dataset for tiny object detection in aerial images, which comes with 700 621 object instances for airplane, bridge, storage tank, ship, swimming pool, vehicle, person, and windmill across 28 036 aerial images. The mean size of objects in AI-TOD is about 12.8 pixels, which is much smaller than others. In AI-TOD-v2 [24], the missing annotation and location error problems are considerably alleviated, facilitating more reliable training and validation processes.

b) Object detection based on OBB: Since most ground objects in the overhead view of remotely sensed imagery are directional, research teams in the field have begun to use oriented rectangles for object annotation in order to represent the objects more accurately. Representative datasets in recent years are mainly as follows.

FAIR1M [25], published by the AIRI of the Chinese Academy of Sciences in 2022, is a large-scale fine-grained object detection dataset for remote sensing objects, containing 42 796 annotated images with five main categories of objects: airplane, ship, vehicle, court, and road, which are further subdivided into Boeing737, Boeing747, Boeing787, C919, A220, A321, A330, A350, ARJ21, passenger ship, motorboat,

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fishing boat, tugboat, engineering ship, liquid cargo ship, dry cargo ship, warship, small car (SC), bus, cargo truck, dump truck, van, trailer, tractor, excavator, truck–tractor, basketball court, tennis court, football field, baseball field, intersection, roundabout, bridge, and other 37 types of fine-grained objects, more than one million instances; the spatial resolution of the image is 0.3 m∼0.8 m; and the scale of the image is minimum 1000 × 1000 and maximum 10 000 × 10 000, which contains a large number of small and sparse objects.

The DOTA is a large-scale dataset for object detection in aerial images, published by Wuhan University in 2018. The images are collected from different sensors and platforms. Each image is of size in the range from 800 × 800 to 20 000 × 20 000 pixels and contains objects exhibiting a wide variety of scales, orientations, and shapes. The instances in DOTA images are annotated by experts in aerial image interpretation by arbitrary (8 d.o.f.) quadrilateral. It has three versions; DOTA-v1.0 [26] contains 15 common categories, 2806 images, and 188, 282 instances. DOTA-v1.5 [27] uses the same images as DOTA-v1.0, but the extremely small instances (less than 10 pixels) are also annotated. It contains 403 318 instances in total. DOTA-v2.0 [28] collects more Google Earth, GF-2 Satellite, and aerial images. There are 18 common categories, 11 268 images, and 1 793 658 instances in DOTA-v2.0. The DOTA is split into training, validation, test-dev, and test-challenge sets. Training contains 1830 images and 268 627 instances. Validation contains 593 images and 81 048 instances. Test-dev contains 2792 images and 353 346 instances. Test-challenge contains 6053 images and 1 090 637 instances.

The SODA-A [29] dataset released by Northwestern Polytechnical University in 2020 is a benchmark dataset specialized for small object detection tasks under aerial scenes, which has 872 069 instances with oriented rectangle box annotation of the airplane, helicopter, small vehicle, large vehicle, ship, container, storage tank, swimming pool, and windmill. It contains 2513 high-resolution images extracted from Google Earth. The average absolute size of instances is only 14.75 pixels.

2) Satellite Video Object Detection Dataset: Compared to static remote sensing images, satellite video mainly focuses on how to fuse the spatiotemporal features between frames to enable effective detection. At present, the object detection datasets in the field of satellite video are relatively small, and the data scale and object categories are more disparate than those of remote sensing image detection datasets, mainly for moving cars. Representative datasets in recent years are given as follows.

The VISO [18] released by the National University of Defense Technology in 2022 is a moving object detection and tracking dataset containing 47 video sequences captured by the Jilin-1 satellite with frame-by-frame annotation of vehicles, aircraft, ships, and trains, for a total of 1 646 038 annotated instances. The dataset is annotated with moving objects only, and more than 90% of the objects are vehicles.

In 2020, the University of New South Wales produced a vehicle detection dataset based on Sky-sat satellite data acquired in the Las Vegas region of the United States [30]. Two proximity regions of 400 × 400 pixels and 600 × 400 pixels were extracted from the same video, each with 700 frames, containing a total of 80 278 vehicle instances.

In 2019, Fudan University published a moving vehicle object detection dataset [31] based on the satellite video of the Valencia region captured by Jilin-1. It includes three video segments of the Valencia region acquired by the Jilin-1 satellite, containing 33 640 frame-by-frame annotated moving vehicle objects.

B. Satellite Video Object Tracking Dataset

Previous studies have shown that there are several datasets available for satellite video object tracking that allows a fair and standardized evaluation of tracking algorithms.

The SatSOT dataset [17], released by the Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences, in 2022, is a dataset focused on satellite video single-object tracking. It contains 105 video sequences with 27 664 frames from three commercial satellite sources: Jilin-1, Skybox, and Carbonite-2. The object categories include cars, planes, ships, and trains. The original scenes in the dataset have a resolution of 12 000 × 5000 pixels, and the video frame rates are 10–25 frames/s (FPS).

As mentioned above, the VISO dataset [18] also supports object tracking in satellite video, providing 3159 sequences for the single-object tracking task and 47 sequences with 3711 individual instances for the multiobject tracking task, with object categories of airplane, car, ship, and train. The original scenes in the dataset have a resolution of 12 000 × 5000 pixels, and the frame rate of the videos is 10 FPS.

The AIR-MOT [7] dataset, published by the Aerospace Information Research Institute of the Chinese Academy of Sciences in 2022, contains ten complete scenes with 149 videos collected by the Jilin-1 satellite. The dataset is labeled with 5736 instances using axis-aligned bounding boxes and contains both airplane and ship objects. Each video has a frame rate of 5–10 FPS and a resolution of 1920 × 1080 pixels.

C. Satellite Video Object Segmentation Dataset

Video object segmentation is a higher level task that combines object detection and semantic segmentation. In the aerospace field, video object segmentation based on satellite overhead views has received more attention.

The data collection methods of the remote sensing object segmentation dataset and general image segmentation dataset have obvious differences: the data sources of the general image segmentation dataset are abundant, mostly handheld cameras, vehicle-mounted cameras, surveillance cameras, and so on, and the viewpoints are mostly flat viewpoints with large object scale and small scene scale. Remote sensing data are generally bird's-eye viewpoints, which can obtain information about larger areas and have greater utilization value in the field of land use. The satellite remote sensing data source has a higher image height and contains more individual objects, but the individual object information is more blurred, and the object category is significantly smaller than the general field. The video object segmentation dataset can continuously observe the object area within a certain area, which has a more

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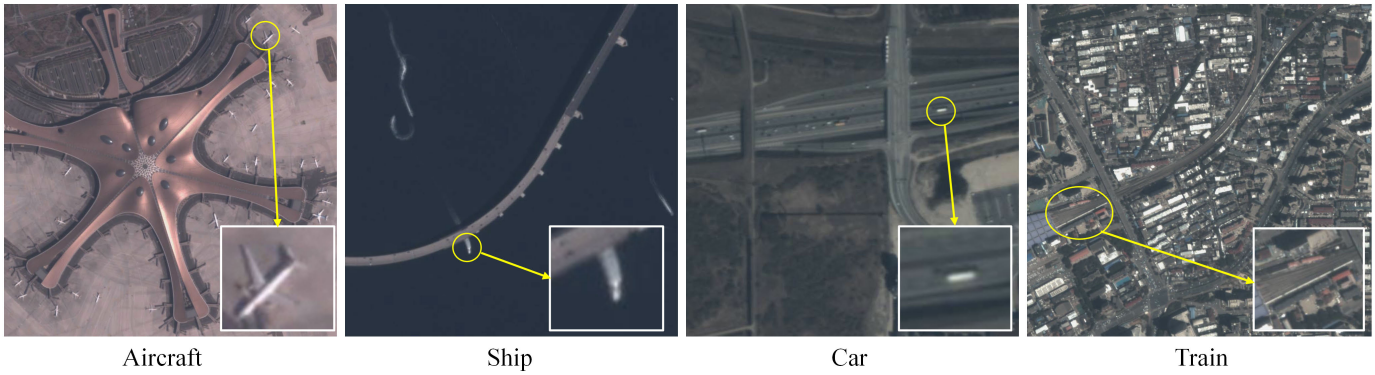


Fig. 1. Examples of video frames containing aircraft, ships, cars, and trains.

important application value in traffic and security. The remote sensing object segmentation dataset has the problems of small object size, less texture information, the large influence of illumination, background interference, and so on compared with the general segmentation dataset. Representative datasets are given as follows.

The SatSOS dataset, published by the Technology and Engineering Center for Space Utilization, Chinese Academy of Sciences, in 2022, is a single-object segmentation dataset based on satellite video, containing six types of categories: aircraft, train, ship, car, building, and cloud. There are 125 objects with 15 000 frames annotated at the pixel level. The dataset focuses on the motion of typical objects on the ground and covers the problems of sensor jitter, object occlusion, and object rotation that will be encountered in video object segmentation, as well as the problems of weak object texture, inconspicuous features, and strong illumination changes that exist in satellite video, providing a benchmark for the research of related segmentation methods.

The iSAID [32] dataset released by Wuhan University in 2019 contains 15 classes with a total of 655 451 instances and 2806 images, with up to 8000 instances and an average of 239 instances for a single image, which fully reflects the common features and scale distribution differences of remote sensing images, making it the first large-scale instance segmentation dataset in remote sensing.

III. Proposed Dataset

A. Data Collection and Annotation

1) Data Source and Scene Selection: The SAT-MTB dataset was collected from various regions of the world by the Jinlin-1 03 satellites, including China, USA, Mexico, France, Italy, Spain, Australia, Japan, Turkey, India, Thailand, Saudi Arabia, and United Arab Emirates (UAE), from 2017 to 2021. All image sequences are represented in true color, corresponding to spectral channels B1: 580∼723 nm, B2: 489∼585 nm, and B3: 437∼512 nm, with a spatial resolution of 0.92 m at the sub-satellite point. The video covers an area size of 11 km × 4.6 km, and the frame rate is 10 FPS.

All raw data were meticulously selected at full resolution, and finally, 249 videos were created to cover the maximum number of object instances possible. These videos contain sequential frames of terrain scenes, such as airports, harbors, roads, bridges, railway stations, and lakes, and incorporate different illumination and weather conditions. The typical objects, such as aircraft, ship, car, and train, in various states of high-speed movement, slow movement, turning, stationary, occlusion, in or out of the field of view, and so on, make the dataset more challenging for multiple tasks. In addition, each scene region was cropped to an optimal size to cover as many types of objects and states of motion as possible, ensuring the diversity and richness of the dataset. Examples of video frames containing aircraft, ships, cars, and trains are shown in Fig. 1.

2) Classification Criteria and Multitask Annotation: We carefully analyzed both the appearance and movement characteristics of objects in each video to assess intraclass distinguishability. Combining with common classification criteria for fine-grained objects, we finally designed two-level category criteria. The first level consisted of four coarse-grained categories: aircraft, ship, car, and train. The second level consisted of 14 fine-grained categories: wide-bodied aircraft (WA), narrow-bodied aircraft (NA), rear-engined aircraft (RA), four-engine aircraft (FA), corporate aircraft (CA), speed boat (SB), yacht (YH), cruise (CS), freighter (FH), naval vessel (NV), other ship (OS), large car (LC), SC, and train (TN). Examples of fine-grained labeled object categories are shown in Fig. 2.

Based on the defined classification criteria, we annotated all instances in each frame of the satellite video for three tasks: object detection, object tracking, and object segmentation. For object detection, we provide both HBB and OBB annotations, as shown in Fig. 3. The annotation file records the sequence number of each frame, the identification number of the same object among frames, and so on. According to the definition of video object detection, all objects of interest that appear in each frame need to be detected. Considering the small size and weak appearance of cars in satellite videos, the object detection task does not include such objects.

For object tracking, we utilize a standard annotation format commonly used in the domain. The annotation records the frame sequence number, object number, coordinate of the upper left corner, width, and height of the bounding box, score, coarse-grained category, fine-grained category, and information about whether the object is truncated or occluded and

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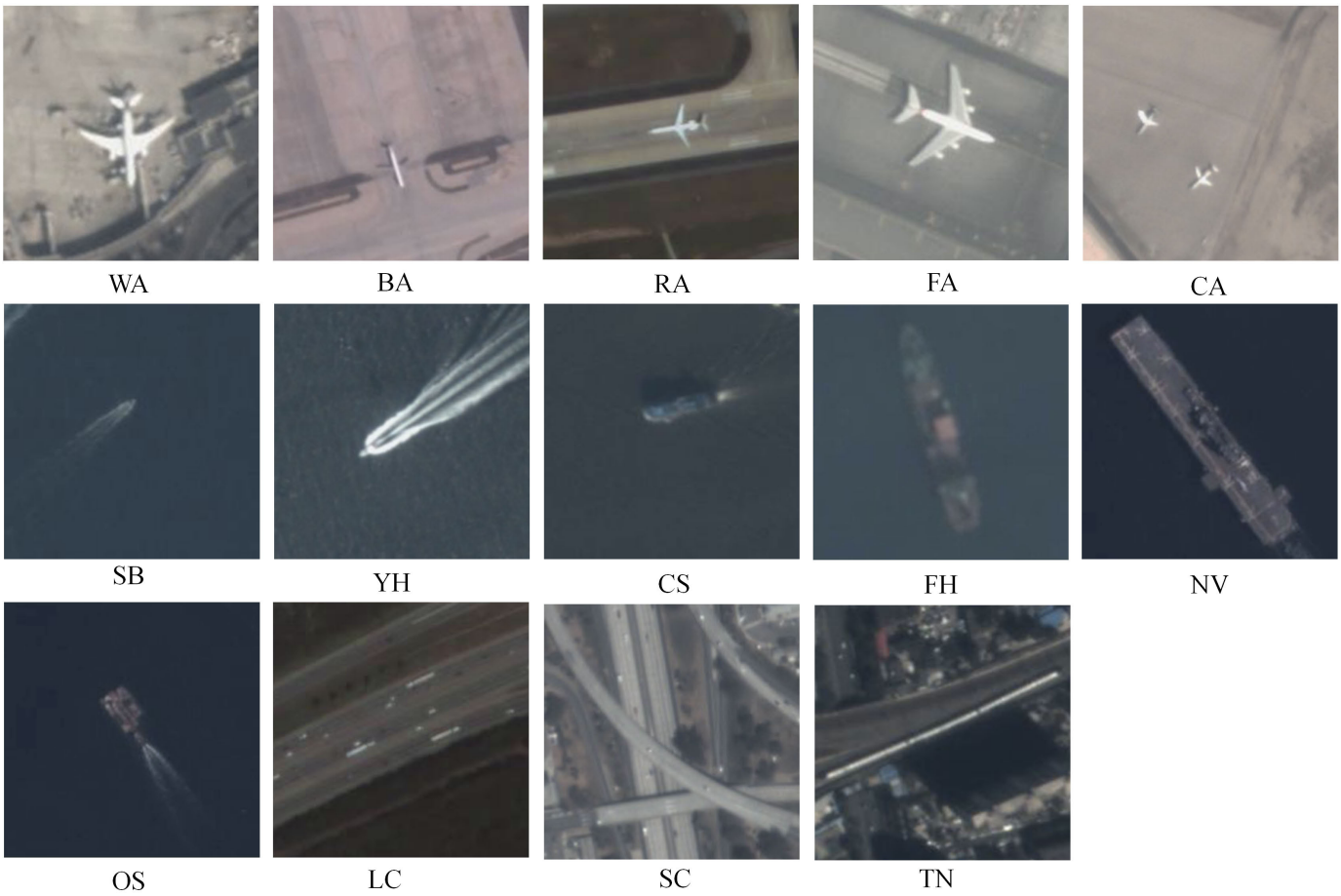


Fig. 2. Example of fine-grained object categories.

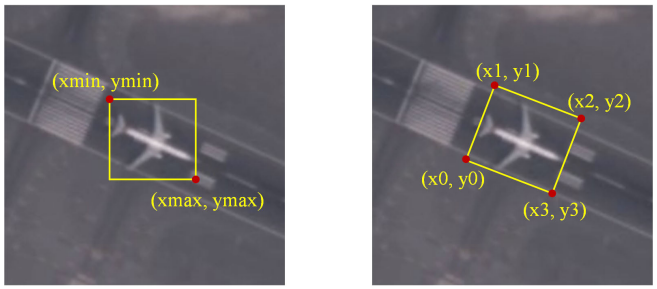


Fig. 3. Example of object detection annotation boxes.

so on. Unlike object detection, tracking task only focuses on moving objects. Static objects in the video are not annotated. It is worth noting that, based on object tracking annotations, it can generate annotations for detecting 14 categories of moving objects, including cars.

Consistent with the object detection task, the object segmentation task also provides annotation of all objects of interest, which appear in each frame of the satellite video, regardless of whether they are in motion or not. The object segmentation task uses a mask to represent the boundary areas of each instance. Each frame of the satellite video corresponds to an annotation file. The annotation information can also be indexed to obtain the corresponding frame number in video sequences and uniquely identify the same object across different frames.

TABLE I
STATISTICS FOR TASKS AND OBJECT SCENARIOS

Scenario	Detection		Tracking	Segmentation
	HBB	OBB		
Aircraft	✓	✓	✓	✓
Ship	✓	✓	✓	✓
Car	–	–	✓	–
Train	✓	✓	✓	✓

Video scenarios containing aircraft, ships, and trains in the dataset are labeled for multiple tasks, such as HBB detection, OBB detection, moving object tracking, and object segmentation, while video scenarios containing cars are labeled for moving object tracking tasks. The statistics of tasks and object scenarios are shown in Table I, and examples of visualization of labeling results are shown in Figs. 4–6.

B. Statistics

1) Statistical Overview: The statistical results of the dataset for each task and annotation format are shown in Table II.

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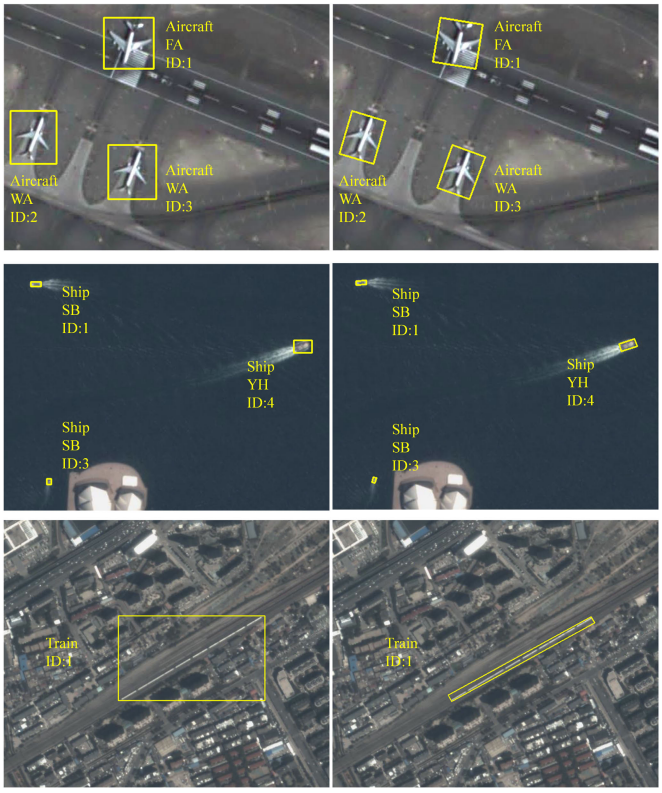


Fig. 4. Example of visualization of annotation results for object detection. (Left) HBB. (Right) OBB.

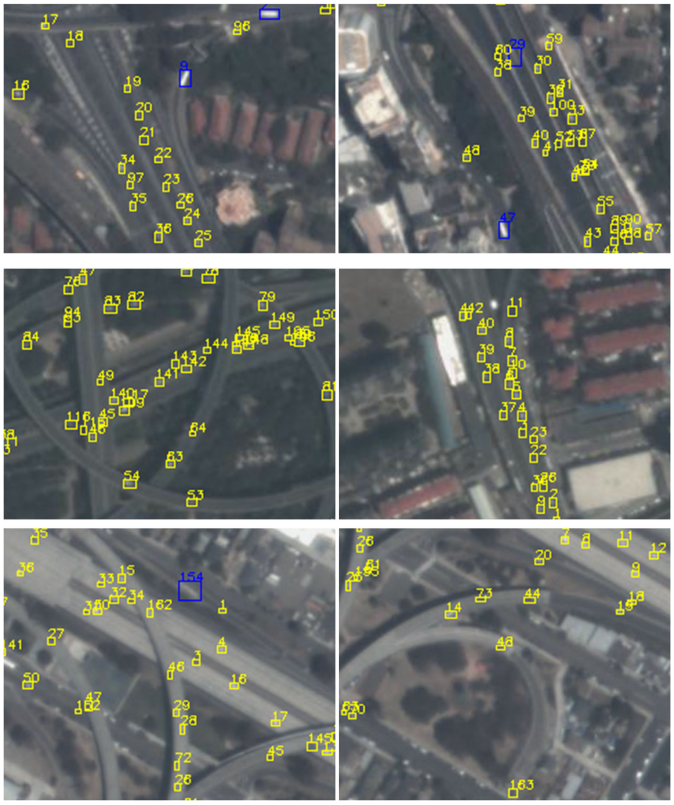


Fig. 5. Example of visualization of annotation results for object tracking (yellow: SC; blue: LC).

We also compare critical indicators with peer datasets in Table III.

2) Statistical Analysis: Due to the differences in spatial distribution and range of motion of aircraft, ships, cars, and

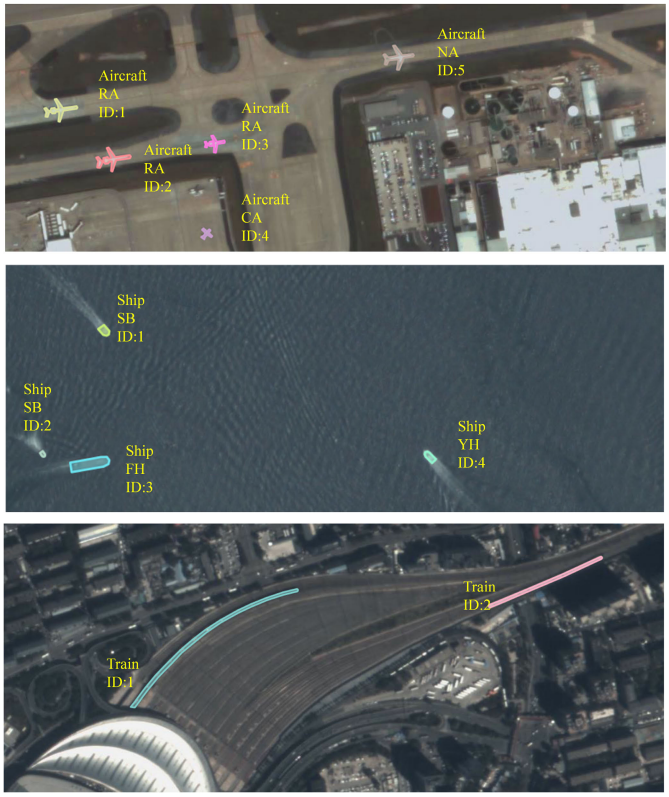


Fig. 6. Example of visualization of annotation results for object segmentation.

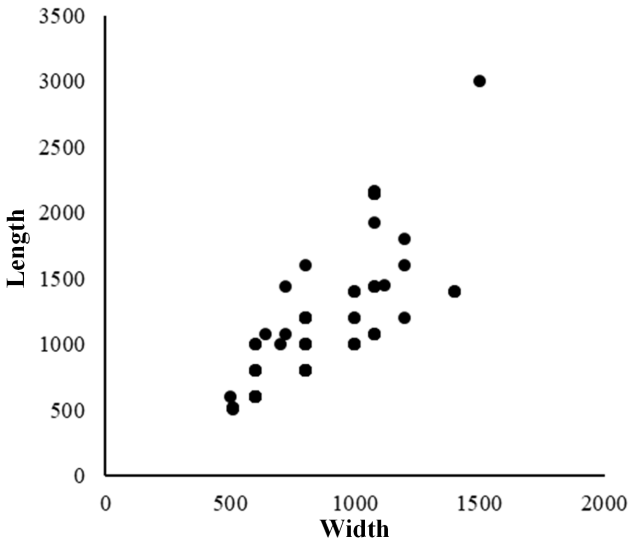


Fig. 7. Absolute pixel size statistics of videos in the dataset.

trains in original videos, the video sizes of the dataset are not uniform to maximize the diversity of categories, motion states, object orientations, and backgrounds. For example, videos containing trains have a larger size due to their inherent scale, while videos corresponding to cars are smaller in size and densely distributed in localized areas, such as roads and bridges. The statistics of video sizes in the dataset are shown in Fig. 7. Moreover, we also show categorywise scale distribution and absolute size of instances in HBB annotations in Fig. 8 and Table IV.

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TABLE II
VOLUME STATISTICS IN THE SAT-MTB DATASET

Detection	HBB				
	Videos	Frames	Coarse-grained	Fine-grained	Instances
	144	33,228	3	12	308,204
	OBB				
	Videos	Frames	Coarse-grained	Fine-grained	Instances
	106	22,767	3	12	211,425
Tracking	HBB				
	Videos	Frames	Coarse-grained	Fine-grained	Instances
	249	50,046	4	14	1,033,511
	OBB				
	Videos	Frames	Coarse-grained	Fine-grained	Instances
	106	22,767	3	12	205,959
Segmentation	Mask				
	Videos	Frames	Coarse-grained	Fine-grained	Instances
	144	33,228	3	12	308,240

TABLE III
QUANTITATIVE STATISTICS FOR EACH DATASET

	Images	Instance	Classes	Annotation	Sequential	Task
VISO [18]	17,730	1,646,038	4	HBB	YES	Detection+Tracking
FAIR1M [25]	42,796	1,000,000	37	OBB	NO	Detection
DIOR [19], [20]	23,463	190,288	20	HBB/OBB	NO	Detection
HRRSD [21]	21,761	55,740	13	HBB	NO	Detection
Las Vagas [30]	1,400	80,278	1	HBB	YES	Detection
DOTA [28]	2,806	188,282	18	HBB/OBB	NO	Detection
AI-TOD [24]	28,036	700,621	8	HBB	NO	Detection
SODA-A [29]	2,513	872,069	9	OBB	NO	Detection
LEVIIR [22]	22,000	11,000	3	HBB	NO	Detection
Valencia [31]	1,740	33,640	1	HBB	YES	Detection
SatSOT [17]	27,664	105	4	HBB	YES	Tracking
AIR-MOT [33]	24,500	5,736	2	HBB	YES	Tracking
SatSOS	15,000	120	6	Mask	YES	Segmentation
iSAID [32]	2,806	655,451	15	Mask	NO	Segmentation
Ours	50,046	1,033,511	14	HBB/OBB/Mask	YES	Detection+Tracking+Segmentation

TABLE IV
CATEGORYWISE STATISTICS OF ABSOLUTE PIXEL SIZE OF INSTANCES IN MINIMUM (MIN), MAXIMUM (MAX), AND AVERAGE (AVG)

	WA	NA	RA	FA	CA
Min	1,271	321	626	596	93
Max	12,786	6,683	4,687	13,541	1,711
Avg	5,175	1,867	1,908	6,953	574
	SB	YC	CS	FH	NV
Min	10	28	244	78	363
Max	1,105	3,179	4,144	10,906	32,861
Avg	95	433	1,112	3,579	5,816
	OS	BC	SC	TN	
Min	43	10	4	500	
Max	3,560	375	244	24,1374	
Avg	1,206	93	23	44,749	

3) Split of the Dataset: The SAT-MTB dataset is divided into a training set and a validation set in a 6:4 ratio based on the number of videos corresponding to aircraft, ships, cars, and trains, as well as the distribution of fine-grained objects. It is worth noting that object detection, object tracking, and object segmentation all follow consistent data distribution, providing a reliable benchmark for comprehensive analysis and algorithm evaluation across multiple tasks. The training set contains 37, 42, six, and 62 satellite videos of aircraft, ships, cars, and trains, respectively. The validation set includes 25, 28, four, and 30 satellite videos. The statistical distribution of the number of fine-grained object instances in the training and validation sets is shown in Fig. 9. The dataset is publicly available for download at http://www.csu.cas.cn/gb/kybm/sjlyzx/gcxx_sjj/sjj_wxxl/.

C. Characteristics of the Dataset

The dataset that we build in this article has distinctive task characteristics and advantages, providing fundamental data support for various visual tasks on satellite video. Moreover, it also presents novel challenges for researchers in the field.

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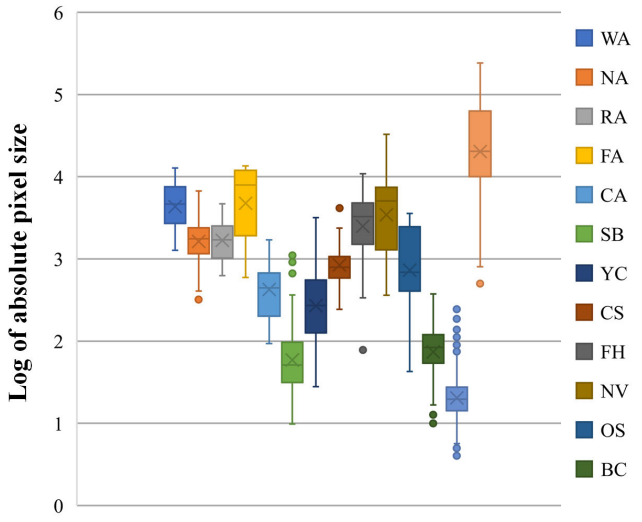


Fig. 8. Categorywise scale distribution of instances in HBB annotations.

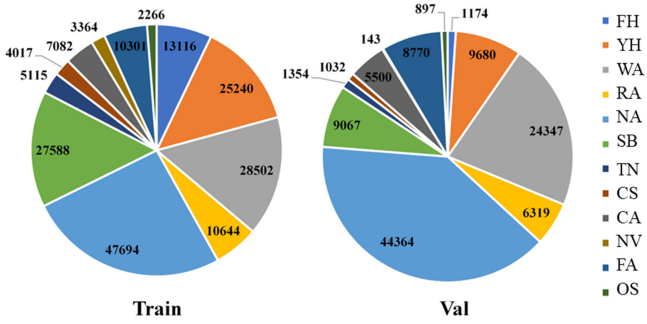


Fig. 9. Statistics on the number of instances of fine-grained objects in the training and validation sets.

The characteristics of the dataset are summarized in the following.

1) Multiple Tasks: This dataset is the first to offer data annotation for three visual tasks: object detection, object tracking, and object segmentation in the same scenario. The multitask dataset provides rich annotations for research in various directions within the field and favorable experimental data conditions for satellite video algorithm development.

2) Large Scale: To the best of our knowledge, this is the largest publicly available annotated dataset in satellite video, supporting three task types, with a diverse range of video scenarios. The dataset includes 249 videos from different regions, a large number of video frames over 50 000, and a large number of annotated instances, totaling 1 033 511. The abundance of annotated data supports the high-quality data samples required for model training.

3) Fine-Grained: This dataset is the first to offer fine-grained category labeling for satellite video. For three basic object categories of aircraft, ships, and cars, it is further refined into five classes of aircraft, six classes of ships, and two classes of cars based on the distinguishability of appearance and motion characteristics. The fine-grained category provides benchmark data for more challenging tasks, such as fine detection, identification, and tracking.

4) Orientation: For satellite video object detection and tracking, we innovatively propose the first OBB annotation in the field of satellite video, which can provide data support for the subsequent network design and experimental evaluation of oriented objects in the satellite video.

IV. Evaluation of the Dataset

A. Object Detection

1) Survey of Object Detection Methods:

a) Object detection of remote sensing imagery based on HBB: Depending on whether a predefined anchor is required, deep learning-based image object detection methods can be classified as anchor-based methods [8], [9], [34], [35], [36], [37] (single-stage and two-stage) and anchor-free [38], [39] methods. The anchor-based methods model the object detection task as a problem of classification and regression of candidate regions, where the candidate regions are generated by sliding windows in single-stage detection and by region generation networks in two-stage detection. Anchor-based methods are computationally complex due to a large number of anchors that introduce many hyperparameters. Subsequent methods have significantly reduced the number of network hyperparameters by identifying key points for object detection [38]. Recently, Transformer has achieved great success in the field of natural language processing (NLP). Inspired by the powerful relational modeling representation capabilities of the Transformer, researchers have proposed to adapt the Transformer to object detection tasks in remote sensing images [39], [40]. Compared to conventional natural images, remote sensing images have their unique characteristics, so researchers focus on the specific problems of remote sensing images to develop algorithms. R2-CNN [41] extracts feature through a lightweight backbone and filters out subimage blocks that do not contain objects by judgmental localization strategy. To address the problem of large variations in object scales of remote sensing images, Zhang et al. [42] employed a multiscale candidate region network and multiscale detectors to detect objects at different scales. To solve the problem of failure to detect small objects in remote sensing images, YOLT [43] is proposed by increasing the upsampling, mapping the intermediate shallow features to the output, and fusing multiple scales to find smaller objects. Long et al. [44] proposed an unsupervised fraction-based bounding box regression and combined it with a nonmaximum suppression to improve the location accuracy of densely distributed objects. Li et al. [45] designed a saliency pyramid fusion strategy in the feature map fusion process to weaken the influence of background and enhance the feature representation of objects. They also combine global attention with the complex background of remote sensing images.

b) Object detection of remote sensing imagery based on OBB: Most objects in remote sensing imagery are directional, and OBB can provide a more accurate contour than HBB. For object detection of remote sensing imagery based on OBB, calculating object angles is one of the main challenges. There

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Fig. 10. Example of visualization results of object detection (yellow: detection results of the DERT; green: ground truth).

are two common solutions for representing OBB, namely, the five-parameter method and the eight-parameter method [46]. The five-parameter method consists of the coordinates of the center of the object, the width, height, and the object angle representing the rotating rectangle external to the object, where the period of the angle is usually chosen as 180° or 90°. The eight-parameter method includes information about the coordinates of the four corners of the OBB, allowing more flexible regression of any quadrilateral, but also results in too many parameters. Most existing frameworks for OBB object detection are mainly improved by the above parameter representations. Xia et al. [26] introduced new regression parameters in Faster R-CNN based on the above parameter representation solution to detect rotating boxes. Ma et al. [46] designed RRPN that regresses the rotation box of objects by manually defining an anchor box with angles while normalizing the features during detection using a designed rotation feature pooling operation, enabling the detection and representation of oriented objects. Jiang et al. [47] proposed R2CNN to generate rotated candidate regions by the region proposal network (RPN) and use multiscale pooling operation to enhance the generalization of detection recognition features. Ding et al. [27] proposed the RoI Transformer that converts the horizontal region generated by the RPN into a rotated region by inserting a lightweight module between RPN and RCNN to minimize the computational complexity associated with rotating object detection. To further improve the feature representation, Cao et al. [48] regressed the HBB and OBB simultaneously to improve each other's accuracy.

c) Satellite video object detection: Compared to image-based object detection, the most important characteristic of video object detection is the addition of contextual information, where each frame has a contextual connection correspondence and similarity. The detection accuracy of the current frame can be improved by using the detection results of the previous and subsequent frames because there is a subcontextual relationship. Since the previous and subsequent frames have similar continuity, redundant information can be used to speed up the detection of each frame. Current video object detection based on deep learning can be divided into algorithms that combine detection, tracking, and algorithms that are based on motion information. For the former, the object is detected for each frame in the video, and then, the object frame is tracked using the tracking algorithm, with the results of the tracking used to correct the previous detection results. Representative algorithms include T-CNN [49], which combines the temporal information learned from the video object tracking task with the spatial features learned from the static object detection task to improve the performance of the video object detection task. Motion information-based video object detection algorithms use the change of pixels in a sequence of images in the time domain and the correlation between adjacent frames to find the correspondence that exists between the previous frame and the current frame. FAGA [50] extracts the optical flow information of the video based on FlowNet [51] and fuses the features of adjacent frames according to the extracted optical flow, enhancing the discriminative of the features. Association LSTM [52] is an online video object detection algorithm using motion information, which mainly consists of the static image object detection algorithm SSD [53] and LSTM [54]. The features of the object are extracted by stacking the SSD detection results, and the spatiotemporal consistency of the object is maintained using the memory property of the LSTM. STMM [55] is a bidirectional recurrent convolutional network that obtains features of spatial characteristics by performing a convolutional stack on multiple adjacent consecutive frames, with the output sent to the classification and regression subnetworks. The algorithm can make use of information about the motion of the object between adjacent frames and also learn information about the motion and appearance changes of the object over long time periods.

Current satellite video-based object detection is mainly based on traditional algorithms, such as background modeling, and mainly focuses on moving vehicles. Research on satellite video object detection based on deep learning is still in its infancy. Jie et al. [56] proposed a cross-frame key point detection network (CKDNet) and designed a cross-frame module to assist in the detection of key points, making full use of the complementary information between frames. Not only the

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TABLE V
COMPARISON OF BENCHMARK EVALUATION RESULTS FOR IMAGERY BASED ON HBB

	Cascade RCNN [34]	Faster RCNN [8]	Libra RCNN [35]	RetinaNet [9]	ATSS [36]	YOLO [37]	FCOS [38]	DETR [39]
Aircraft	0.678	0.688	0.619	0.623	0.688	0.604	0.610	0.637
Ship	0.362	0.400	0.393	0.298	0.399	0.232	0.252	0.379
Train	0.551	0.428	0.524	0.420	0.377	0.442	0.402	0.489
mAP	0.530	0.505	0.512	0.447	0.488	0.426	0.421	0.502
WA	0.515	0.522	0.479	0.488	0.572	0.505	0.527	0.508
NA	0.587	0.620	0.430	0.492	0.560	0.488	0.433	0.545
RA	0.616	0.649	0.479	0.581	0.546	0.607	0.376	0.639
FA	0.264	0.366	0.299	0.233	0.301	0.497	0.256	0.479
CA	0.217	0.208	0.138	0.105	0.181	0.302	0.125	0.372
SB	0.004	0.007	0.001	0.085	0.034	0.088	0.059	0.066
YH	0.283	0.321	0.336	0.240	0.330	0.294	0.276	0.369
CS	0.000	0.000	0.000	0.000	0.000	0.004	0.000	0.001
FH	0.248	0.098	0.327	0.091	0.317	0.055	0.096	0.432
NV	0.000	0.000	0.002	0.000	0.000	0.000	0.000	0.000
OS	0.015	0.022	0.089	0.003	0.323	0.657	0.118	0.119
TN	0.551	0.428	0.524	0.420	0.377	0.442	0.360	0.489
mAP	0.275	0.270	0.258	0.228	0.295	0.328	0.219	0.335
FPS	15.7	19.0	17.9	20.2	21.2	39.7	15.9	7.9
FLOPs(G)	234.5	206.7	207.7	209.2	201.9	194.1	197.2	91.6
Params(MB)	68.9	41.2	41.4	36.3	31.9	61.6	31.2	41.2

spatial information of the same frame is extracted but also the relative spatial relationship of neighboring vehicles is taken into account. Xiao et al. [57] proposed a two-stream dynamic and static fusion network (DSFNet) consisting of a 2-D backbone with static contextual information from individual frames and lightweights, and a 3-D backbone that extracts continuous dynamic motion frames. Pi et al. [58] designed a feature interframe differential module to obtain neighboring motion information and extract semantic features. They further introduce Transformer to refine the semantic features for effective vehicle detection in response to insignificant satellite video vehicle object appearance information.

2) Experiment Setup: For the object detection of imagery based on HBB experiments, we selected eight representative methods covering several categories, such as two-stage, one-stage, anchor-free, and Transformer-based, including Faster R-CNN [8], Cascade R-CNN [34], Libra R-CNN [35], RetinaNet [9], ATSS [36], YOLOv3 [37], FCOS [38], and Deformable DETR [39]. The relevant models were trained for 12 epochs with a batch size of 16, a learning rate of 0.02, and a weight decay of $1e-4$, and the performance of the models in the final epoch was evaluated. For the object detection of imagery based on OBB experiments, we choose OBB object detection methods based on classical networks with adaptations, including Rotate RetinaNet [9], Rotate Faster R-CNN [8], Rotate FCOS [38], and dedicated networks specifically designed for rotating box objects in remote sensing images, including RoI Transformer [27], R3Det [59], S2A-Net [60], ReDet [61], and Oriented R-CNN [62], for a total of eight representative baselines. The relevant models were trained for 12 epochs with a batch size of 2, a learning rate of $25e-4$, and a weight decay of $1e-4$, and the performance of the models in the final epoch was evaluated. For the video object detection experiments, we selected four representative video object detection methods, namely, DFF [63], FGFA [50], SELSA [64], and Temporal ROI Align [65]. The relevant models were trained for 7 epochs with a batch size of 2, a learning rate of $25e-4$, and a weight decay of $1e-4$, and the performance of the models in the final epoch was evaluated. To maintain fairness, all of the above methods were conducted on a server with an Intel Xeon E5-2640 v4 CPU with 128-GB RAM and two Nvidia RTX GPUs.

3) Evaluation Metrics: For the evaluation of object detection benchmark algorithms on SAT-MTB, we set Intersection over Union (IoU) to 0.5 and calculate the average precision (AP) and mean AP (mAP) of different detection methods on the coarse-grained and fine-grained dimensions.

The precision evaluation indicator is calculated as follows:

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{1}$$

where TP indicates that the predicted value is the same as the true annotation and the predicted value is a positive sample, and FP indicates that the predicted value is different from the true annotation, and the predicted value is a positive sample. MAP calculates the average of all object detection accuracies under the coarse-grained and fine-grained object classification criteria, respectively. It should be noted that training and testing are performed based on the fine-grained labels of SAT-MTB, and coarse-grained results are calculated based on inference from the fine-grained results. In addition, we also evaluate the computational efficiency of different algorithmic models: FPS, floating-point operations (FLOPs), and Params (model parametric quantities), to provide a comprehensive assessment.

4) Results and Analysis:

a) Object detection results for imagery based on HBB: Table V shows the evaluation results of eight benchmark object detection methods for imagery based on HBB, with the precision, mAP, model effectiveness, and parameter metrics for each category of objects under coarse and fine granularity, respectively. A comparison between the DETR detection results and the ground truth is shown in Fig. 10.

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TABLE VI
COMPARISON OF BENCHMARK EVALUATION RESULTS FOR IMAGERY BASED ON OBB

	RRetinaNet [9]	RFaster-RCNN [8]	RFCOS [38]	RoI Transformer [27]	R3Det [59]	S2A [60]	ReD [61]	ORC [62]
Aircraft	0.921	0.931	0.920	0.905	0.893	0.931	0.945	0.913
Ship	0.495	0.499	0.431	0.624	0.554	0.421	0.598	0.495
Train	0.000	0.000	0.001	0.011	0.001	0.000	0.007	0.036
mAP	0.472	0.480	0.453	0.458	0.485	0.451	0.452	0.484
WA	0.842	0.772	0.804	0.803	0.857	0.869	0.803	0.804
NA	0.813	0.850	0.829	0.846	0.830	0.853	0.844	0.844
RA	0.792	0.835	0.857	0.827	0.834	0.883	0.851	0.811
FA	0.868	0.799	0.906	0.906	0.865	0.900	0.818	0.900
CA	0.519	0.609	0.426	0.382	0.207	0.572	0.811	0.422
SB	0.010	0.047	0.034	0.104	0.102	0.021	0.026	0.012
YH	0.226	0.376	0.402	0.541	0.316	0.45	0.487	0.438
CS	0.132	0.016	0.001	0.006	0.402	0.235	0.002	0.141
FH	0.058	0.001	0.000	0.008	0.000	0.000	0.000	0.000
NV	0.000	0.000	0.000	0.001	0.000	0.000	0.002	0.003
OS	0.396	0.414	0.570	0.355	0.557	0.369	0.536	0.364
TN	0.000	0.000	0.001	0.011	0.001	0.000	0.007	0.036
mAP	0.388	0.393	0.403	0.399	0.414	0.429	0.432	0.398
FPS	27.6	14.0	31.4	22.5	20.8	24.6	16.6	24.2
FLOPs(G)	405.65	387.08	390.90	401.05	631.90	373.05	126.79	387.34
Params(MB)	36.35	41.14	31.92	55.11	41.83	38.59	31.62	41.14

From the coarse-grained detection results, it can be seen that mAP of two-stage and Transformer-based DETR algorithms, such as Faster R-CNN, Cascade R-CNN, and Libra R-CNN, exceeds 0.5, while mAP of the single-stage and anchor-free FCOS, such as RetinaNet, ATSS, and YOLOv3, is less than 0.5. In terms of specific coarse-grained categories, due to the more regular appearance and structural form of aircraft in satellite video compared to ships and trains, aircraft have the highest AP with significant distinguishing features. Due to the large intraclass variation in the appearance and size of ships, especially for small ships in motion, where the overall visual representation of the ship's hull is significantly disturbed by the wake, ships have the lowest AP. In terms of fine-grained detection results, DETR and YOLOv3 perform significantly better than the other methods, and the mAP advantage of the two-stage methods in coarse-grained detection is eliminated. The reason for this analysis is that the two-stage methods generate regions of interest via RPN, which can more accurately determine categories for coarse-grained objects with significant interclass differences, whereas the model tends to confuse the fine-grained categories when further discrimination of fine-grained objects is required for similar features, leading to a reduction in mAP. Deformable DETR achieves the highest mAP of 0.335 for fine-grained detection due to the Transformer's powerful learning capability in global modeling, but its computational speed is at the lowest level and cannot meet the demand for real-time detection at satellite video frame rates. YOLOv3 performs better in terms of mAP and efficiency. In terms of specific fine-grained objects, all methods performed better in detecting WA, NA, RA, and so on. This is mainly due to the fact that these objects are highly distinguishable in terms of detail structure pattern and size, and the number of samples in the dataset is larger than in other categories, which makes the trained models perform better in the generalizing of these fine-grained objects. Similar to the coarse-grained performance, the detection results of each method for fine-grained objects under the category of ships are worse than the performance of fine-grained aircraft, with lower AP for all categories except YH and FH, and even for CS and NV where the methods fail, mainly due to the limited samples and the large variation of the samples in appearance features, making it difficult to train the model to acquire the ability to discriminate the above objects.

b) Object detection results for imagery based on OBB: Table VI shows the evaluation results of eight benchmark object detection algorithms for imagery based on OBB, with the precision, mAP, model effectiveness, and parameter metrics for each category of objects under coarse and fine granularity, respectively. From the coarse-grained detection results, it can be seen that the benchmark methods perform best on the aircraft and slightly worse on the ship. The reasons are consistent with the analysis of object detection based on HBB, where aircraft has prominent and significant visual features of appearance and distinguishable structure, and is relatively large in overall size, so all achieve better AP. Ships are smaller in size, have large intraclass variation, and have significant wake due to motion that interferes with the network detecting the features of their subject objects. For trains, any benchmark detection method based on OBB fails because the trains are distributed in lines, the features are sparse, and the aspect ratio is extreme, which makes it difficult to jointly obtain good localization information of objects with relatively normal size and shape and trains. In addition, the distribution of the actual scenes of the trains is limited by the small number of samples, which also makes it difficult to maintain the network to obtain a robust model. The failure to detect the train also results in the benchmark based on OBB with mAP of less than 0.5, which is inferior to the benchmark based on HBB, such as two-stage and Transformer, but the overall mAP performance is more balanced, with an mAP difference of only about 3%. The reason for this is that these two types of objects in satellite video scenes have obvious directionality, and the

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samples are mainly collected in airports, ports, and other areas, which makes it difficult for HBB detection methods to perform tight and precise positioning.

In terms of fine-grained object detection performance, ReDet has the best precision, with mAP of 0.432, while RRetinaNet has the lowest precision, with mAP of 0.388. The overall difference between the two is only 4%, and the difference between the benchmarks for HBB detection is more than 10%, making the performance of the benchmarks based on OBB more balanced. However, the performance of certain fine-grained object categories, such as CA and CS, varies greatly between methods, with an AP difference of up to 40% or more. This is due to differences in the structural design of the different methods, resulting in differences in the ability of each network model to locate and recognize objects of a particular size and shape. For example, for the CS, the R3Det method achieves mAP of 0.402, while the RFCOS is almost ineffective and fails to detect the object. The results also show that, for FH, NV, and SB, the performance of each benchmark is poor. The reason for this is that these objects have a small size and are easily confused with other similar objects, and given that the classification of objects takes into account the appearance and movement of specific objects, it is difficult to rely solely on visual features to distinguish the objects within the class. In terms of detection efficiency, all FPSs of the benchmarks based on OBB exceed the frame rate of the Jilin-1 satellite video, which can all meet real-time detection requirements. However, when comparing the efficiency performance of specific methods, it can be seen that detection accuracy and detection efficiency are mutually exclusive, which is also related to the fact that more computations can achieve more accurate detection.

c) Results of video object detection: Table VII shows the evaluation results of four benchmark algorithms for video object detection, with the precision, mAP, and model effectiveness for each category of objects under coarse and fine granularity, respectively. The coarse-grained object detection results show an 8.3% difference in mAP, with SELSA performing better than the other methods by performing spatiotemporal feature association based on semantic similarity. In terms of specific category performance, aircraft and trains perform better than ships, which is similar to the performance of the HBB image-based object detection methods in Table VII. This indicates that the performance of the algorithms for different schemes depends on the visual characteristics of the objects and the differentiability of the categories, which is also verified in the performance of fine-grained object detection results, such as WA and FA. The accuracy of objects such as WA, FA, and RA is high, while CS, NV, and SB fail. In terms of computational efficiency, the DFF algorithm propagates features from the current frame to the next frame based on optical flow information, avoiding repeated extraction of features from each frame, with significant advantages in computational efficiency, while other benchmarks have significantly reduced computational efficiency due to the addition of complex operations, such as feature similarity calculation and feature fusion in the network, which cannot meet the efficiency requirements of real-time satellite video detection. Since the benchmarks for the video object detection methods in this article are HBB, we compared the performance of the results with those of the HBB image-based object detection methods. It is clear from Tables V and VII that the image-based detection algorithms are overall significantly better than the video-based detection algorithms in terms of mAP and computational efficiency. The reason for this analysis is that the current benchmarks for video object detection are designed for general video characteristics, e.g., the DFF algorithm mainly solves the problem of operational efficiency, while the FGFA, SELSA, and Temporal ROI Align methods mainly improve the representation of current frame features through feature similarity or motion feature fusion to solve problems such as motion blur and appearance degradation. While satellite video has high redundancy and slow motion of the object, and the visual features of the object change less in the temporal sequence, the benchmarks in the general field do not apply to satellite video, and therefore, the performance of object detection is poor.

Through the evaluation and analysis of various benchmarks, such as horizontal boxes image-based object detection, rotating boxes image-based object detection, and video object detection, the challenges and improvement directions of SAT-MTB object detection are summarized as follows.

1) Satellite video is significantly different from general-purpose videos, and it is important to build a satellite video-oriented object detection framework. The SAT-MTB dataset will support and promote research in this field.
2) Some fine-grained object categories are limited by the discriminability of the visual representation of the object and the sparse number of objects, leading to the failure of various benchmarks, and how to combine the visual characteristics and motion patterns of the object to achieve effective fine-grained detection is one of the future research directions.
3) Trains, large ships, and other objects are distributed in a line shape, the characteristics of the object are sparse and extreme aspect ratio, and how to design a rotating frame detection network according to the typical ground object of satellite video to obtain more accurate localization information of the object is also one of the future research directions.

B. Object Tracking

1) Survey of Object Tracking Methods: Recently, several methods for object tracking in satellite video have been proposed [1], [5], [66], [67]. However, these methods are mainly for a single object, while multiobject tracking in satellite video is still in the initial research stage. The detection-based tracking method is the mainstream multiobject tracking method, which uses an object detector to mine and detect potential objects in the scene, and then correlates the detection results between frames to obtain tracking trajectories. Some studies focus on the detection of moving objects. D&T [31] provides a vehicle detection algorithm based on local noise modeling to distinguish potential vehicles from noisy patterns through

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TABLE VII
COMPARISON OF BENCHMARK EVALUATION RESULTS OF VIDEO OBJECT DETECTION

	DFF [63]	FGFA [50]	SELSA [64]	Temporal-Roi-Align [65]
Aircraft	0.402	0.402	0.438	0.425
Ship	0.207	0.258	0.241	0.124
Train	0.238	0.345	0.343	0.226
mAP	0.282	0.335	0.341	0.258
WA	0.654	0.687	0.710	0.553
NA	0.364	0.421	0.523	0.354
RA	0.671	0.693	0.651	0.443
FA	0.416	0.420	0.465	0.334
CA	0.007	0.007	0.016	0.000
SB	0.000	0.000	0.000	0.001
YH	0.067	0.051	0.073	0.034
CS	0.031	0.013	0.006	0.046
FH	0.212	0.126	0.199	0.063
NV	0.000	0.022	0.002	0.001
OS	0.289	0.159	0.173	0.031
TN	0.238	0.345	0.343	0.226
mAP	0.246	0.245	0.263	0.174
FPS	38.9	8.6	9.4	3.7

exponential probability distributions. DSFNet [57] proposes a dual-stream network with dynamic and static fusion that extracts static contextual information from single-frame and dynamic motion cues from successive frames and then integrates static and dynamic features to detect moving objects in satellite video. CKDNet-SMTNet [56] performs cross-frame keypoint detection using interframe information and constructs a dual-branch structure containing long- and short-term memories to effectively detect and track dense vehicles. Based on the detectors, SORT [68] and DeepSORT [69] use Kalman filtering with the Hungarian matching algorithm to achieve multiobject tracking. ByteTrack [70] proposes a more efficient data association method based on DeepSORT. Tracktor++ [71] presents a tracking prediction model based on the regression of the previous frame's detection results on the next frame, transforming the detector into a tracker. However, these methods are less efficient due to the two-stage processing, and they face difficulties in the association process when the objects are densely distributed and highly similar.

Some methods perform joint detection and tracking with simultaneous object detection and temporal association. CenterTrack [72] proposes an algorithm that applies a key point detection model to the image sequence and the previous frame detection result, and tracks different objects by associating between key points. JDE [73] and FairMOT [74], on the other hand, use a shared network for both detection feature and ReID feature extraction, and associate the predicted IDs frame by frame. TGraM [7] models multiobject tracking as a graphical information inference process from a multitask learning perspective and presents a graph-based spatiotemporal inference module to mine potential higher order correlations between video frames. Although these single-stage methods save inference time, they are challenging to detect and track well objects that lack appearance information well. There are now some alternative approaches to accomplish multiobject tracking. TubeTK [74] proposes a 3-D pipeline detection and tracking structure that simultaneously covers temporal, spatial, and motion information. SiamMOT [75] incorporates the correlation structure in Siamese tracking for multiobject tracking. In addition, LGM [76] uses graph structures for interobject and interframe correlation. MOTR [77] introduces the Transformer structure to achieve multiobject tracking. We select benchmark methods from the above methods, and train and test these benchmark methods on our dataset to further investigate the significance of each method for multiobject tracking in satellite video.

2) Experiment Setup: First, we selected eight benchmark methods for training, testing, and evaluation comparison, including DeepSORT [69], Tracktor++ [71], DSFNet [57], CenterTrack [72], FairMOT [33], TGraM [7], ByteTrack [70], and CKDNet-SMTNet [56]. We used the YoloX [78] detector to provide the detection results for the DeepSORT and ByteTrack trackers, and the Faster R-CNN [8] detector to provide the detection results for Tracktor++. Note that most of the methods are designed for a single category. To realize the four categories included in the SAT-MTB dataset, we adapted each method to output the four-category results separately. In addition, due to the large size difference between the four categories, we normalized the stride to 2 for all methods with a stride greater than 2 to balance the size difference between the different categories. During the training process, we randomly cropped the training images into small 512 $\times$ 512 patches to unify the input format for training. To maintain fairness, we kept the other details of each tracking method consistent with the original article. For testing, we ran experiments at the original image resolution. All tests were performed on an Intel Xeon E5-2640 v4 CPU with 128-GB RAM and an Nvidia RTX GPU.

3) Evaluation Metrics: The evaluation metrics include multiobject tracking accuracy (MOTA), IDF1 score (IDF1), the

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TABLE VIII
ACCURACY COMPARISON OF CAR CATEGORY FOR MULTIOBJECT TRACKING

Cate	Method	MOTA↑	IDF1↑	MT↑	ML↓	FP↓	FN↓	IDs↓
Car	DeepSORT [69]	29.0%	43.9%	469	1958	6155	237226	707
	CKDNet-SMTNet [56]	27.1%	45.3%	368	735	73022	172994	4808
	Trackor++ [71]	-5.4%	2.0%	9	3117	23763	337778	875
	ByteTrack [70]	27.9%	43.3%	534	2153	3660	244072	102
	DSFNet [57]	48.1%	63.9%	1121	1073	12719	165022	694
	FairMOT [33]	8.5%	18.5%	102	2417	9529	291156	13895
	CenterTrack [72]	24.6%	30.5%	551	1787	11840	242197	5228
	TGraM [7]	4.8%	12.1%	57	2668	8117	309047	10307

percentage of mostly tracked trajectories (MT), mostly lost trajectories (ML), the number of false positives (FPs), false negatives (FNs), and identity switches (IDs). MOTA is the most widely used metric to evaluate the overall performance of multiobject tracking methods. This metric is a combination of FP, FN, and IDs. IDF1 represents the proportion of detected objects with correct IDs among the detected and tracked objects, combining ID accuracy and recall. MT represents the proportion of trajectories correctly tracked in 80% of the frames, while ML represents the proportion of trajectories not correctly tracked in 80% of the frames. FP and FN correspond to the number of false detections and missed detections of the object, respectively, while IDs correspond to the number of ID switches during tracking. MOTA, FP, and FN focus on the detection performance, while IDF1, IDs, MT, and ML focus on the tracking performance.

4) Results and Analysis: In the multiobject tracking task, the IoU threshold for determining successful tracking is generally set at 0.5. However, for small objects, such as cars and ships in satellite video, this threshold can produce results that are significantly different from reality, and some successfully tracked objects will be classified as tracking failures in the evaluation. Therefore, we provide the results under an IoU threshold of 0.3, thus providing a comprehensive evaluation of the multiobject tracking performance of the benchmark methods. As shown in Tables VIII–XI and Fig. 11, the benchmark methods performed best for aircraft, second best for ships and cars, and worst for trains. In general, aircraft have more obvious visual characteristics, and the object size is usually larger. In contrast, cars are small and dense, and ships vary considerably in size between individuals and are susceptible to motion wakes. The narrow shape of train objects makes tracking more difficult, and there are fewer trains in the dataset. For the MOTA metric, DSFNet performs best for cars and aircraft (48.1% and 70.9%), while CenterTrack performs best for ships and trains (54.2% and 6.6%). For both the MT and ML metrics, it can be seen that our dataset poses a greater challenge for car and ship tracking. Even with the lowest ML, CKDNet-SMTNet still fails to track 23.1% of cars and 16.9% of ships. As for the IDs metric, car categories even occur at $1e-5$ orders of magnitude of IDs. ByteTrack has the lowest IDs, while more IDs mean instability of the tracking method. FairMOT and TGraM have the highest IDs, implying that ReID feature extraction is not applicable for car tracking in dense scenes with featureless and extremely similar objects. The combined FP and FN metrics give the detection performance of the tracking methods. It is worth noting that CenterTrack has the best performance for the ships, and CenterTrack also has the best overall performance for four categories, meaning that the method has a solid ability to detect objects with large-scale differences.

However, the performance of these baseline models has much room for improvement, as there is a large amount of ML in car and ship sequences, while almost all train sequences are ML. This means that the existing trackers still need to be improved in terms of simultaneous detection and tracking of multiscale objects, and the robustness of the tracking methods should be further enhanced. Our dataset presents a higher challenge for cars, and the selected scenes focus mainly on highways and bridges with dense object distribution, where objects with similar appearance bring more IDs to the trackers and reduce the overall tracking performance. In addition, most current tracking methods can only handle a single category, and multiple categories of moving objects, multiple complex scenes, and multiple object scales make our dataset more difficult. In summary, satellite video multiobject tracking is challenging, especially when faced with multiclass and multiscale objects, dense distribution of similar objects, and low resolution. Our SAT-MTB dataset will further advance the research in satellite video multiobject tracking.

C. Object Segmentation

1) Survey of Object Segmentation Methods: According to the network framework, video object segmentation methods can be divided into CNN- and memory-based segmentation methods. CNN-based segmentation methods are performed on separate video frames, while memory-based segmentation methods segment the video directly.

a) CNN-based segmentation methods: He et al. [79] proposed Mask R-CNN that is an extension of Faster R-CNN [8]. Mask R-CNN includes mask segmentation branches and incorporates RoIAlign, a method that uses bilinear interpolation to eliminate misalignment between the region of interest (RoI) and the extracted features. Cascade Mask is a novel object detection and segmentation framework that combines the strengths of Cascade R-CNN and Mask R-CNN. In Cascade Mask, an incremental IoU thresholding strategy is employed to generate high-quality bounding boxes and segmentation masks. Huang et al. [80] proposed Mask Scoring

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TABLE IX
ACCURACY COMPARISON OF AIRCRAFT CATEGORY FOR MULTIOBJECT TRACKING

Cate	Method	MOTA↑	IDF1↑	MT↑	ML↓	FP↓	FN↓	IDs↓
Aircraft	DeepSORT [69]	35.6%	55.3%	22	33	661	9400	0
	CKDNet-SMTNet [56]	54.3%	65.9%	53	3	4349	2700	91
	Trackor++ [71]	17.6%	57.8%	33	13	6957	5856	64
	ByteTrack [70]	45.3%	65.1%	31	26	916	7636	1
	DSFNet [57]	70.9%	83.3%	42	7	712	3816	12
	FairMOT [33]	32.0%	63.9%	32	22	4413	6218	1
	CenterTrack [72]	54.9%	71.1%	36	14	1736	5233	82
	TGraM [7]	37.4%	67.5%	33	18	4368	5405	6

TABLE X
ACCURACY COMPARISON OF SHIP CATEGORY FOR MULTIOBJECT TRACKING

Cate	Method	MOTA↑	IDF1↑	MT↑	ML↓	FP↓	FN↓	IDs↓
Ship	DeepSORT [69]	13.5%	45.5%	35	40	6012	12641	144
	CKDNet-SMTNet [56]	36.9%	49.8%	43	14	5403	8025	299
	Trackor++ [71]	34.3%	56.6%	35	30	3466	10695	119
	ByteTrack [70]	10.5%	46.9%	38	34	8054	11233	168
	DSFNet [57]	37.2%	57.7%	38	16	4306	9116	235
	FairMOT [33]	8.2%	45.0%	23	39	6961	12860	143
	CenterTrack [72]	54.2%	61.9%	53	15	3541	6186	238
	TGraM [7]	-13.9%	35.6%	19	47	10103	14383	283

TABLE XI
ACCURACY COMPARISON OF TRAIN CATEGORY FOR MULTIOBJECT TRACKING

Cate	Method	MOTA↑	IDF1↑	MT↑	ML↓	FP↓	FN↓	IDs↓
Train	DeepSORT [69]	-7.1%	0.0%	0	8	132	1851	0
	CKDNet-SMTNet [56]	-1.0%	42.5%	3	2	910	952	7
	Trackor++ [71]	3.6%	20.4%	1	6	288	1488	9
	ByteTrack [70]	0.0%	0.0%	0	8	0	1851	0
	DSFNet [57]	0.0%	0.0%	0	8	0	1851	0
	FairMOT [33]	2.0%	6.9%	0	7	31	1781	2
	CenterTrack [72]	6.6%	23.6%	0	6	152	1574	3
	TGraM [7]	1.1%	3.3%	0	8	10	1814	6

R-CNN by adding MaskIoU head to regress IoU between instance mask and ground truth, and avoid the gap between mask quality and classification confidence score. Chen et al. [81] proposed the hybrid task cascade (HTC) that enhances the performance of the Cascade Mask R-CNN by incorporating mask information flow. The approach involves constructing a joint multistage model that weaves together bounding boxes and mask branches to improve the accuracy of object detection and segmentation. Inspired by the effectiveness of data augmentation (i.e., Crop-and-Paste [82]), Fang et al. [83] proposed InstaBoost that involves separating the foreground from the background using matting [84] and inpainting [85] algorithms and then identifying suitable locations to paste the foreground by considering its appearance similarity. With the goal of improving the coarse edge of segmentation methods, Kirillov et al. [86] introduced PointRend that approaches the segmentation task as an image-rendering process. This method utilizes an adaptive selection process to choose the top N uncertain points and then iteratively predicts these points to produce high-quality and refined masks. Motivated by visual and cognitive processes, Qiao et al. [87] integrated the recursive feature pyramid (RFP) and switchable atrous convolution (SAC) into the HTC architecture to create DetectoRS. The RFP leverages feedback connections from the FPN to bottom-up backbone layers via spatial pyramid pooling, while SAC utilizes atrous convolutions with varying kernel sizes to capture receptive fields at multiple scales. Vu et al. [88] developed the sample consistency network (SCNet) as an extension to the Cascade Mask R-CNN. SCNet introduces a mask branch after the last bounding box regression stage to ensure sample consistency. Fang et al. [89] proposed QueryInst that incorporates query-based detection into instance segmentation tasks. This approach employs a multihead attention mechanism and dynamic convolution to learn instance features from the query. Inspired by one-stage object detection, representative one-stage methods have also emerged for instance segmentation tasks. Bolya et al. [90] proposed a single-shot, anchor-free, real-time instance segmentation method named

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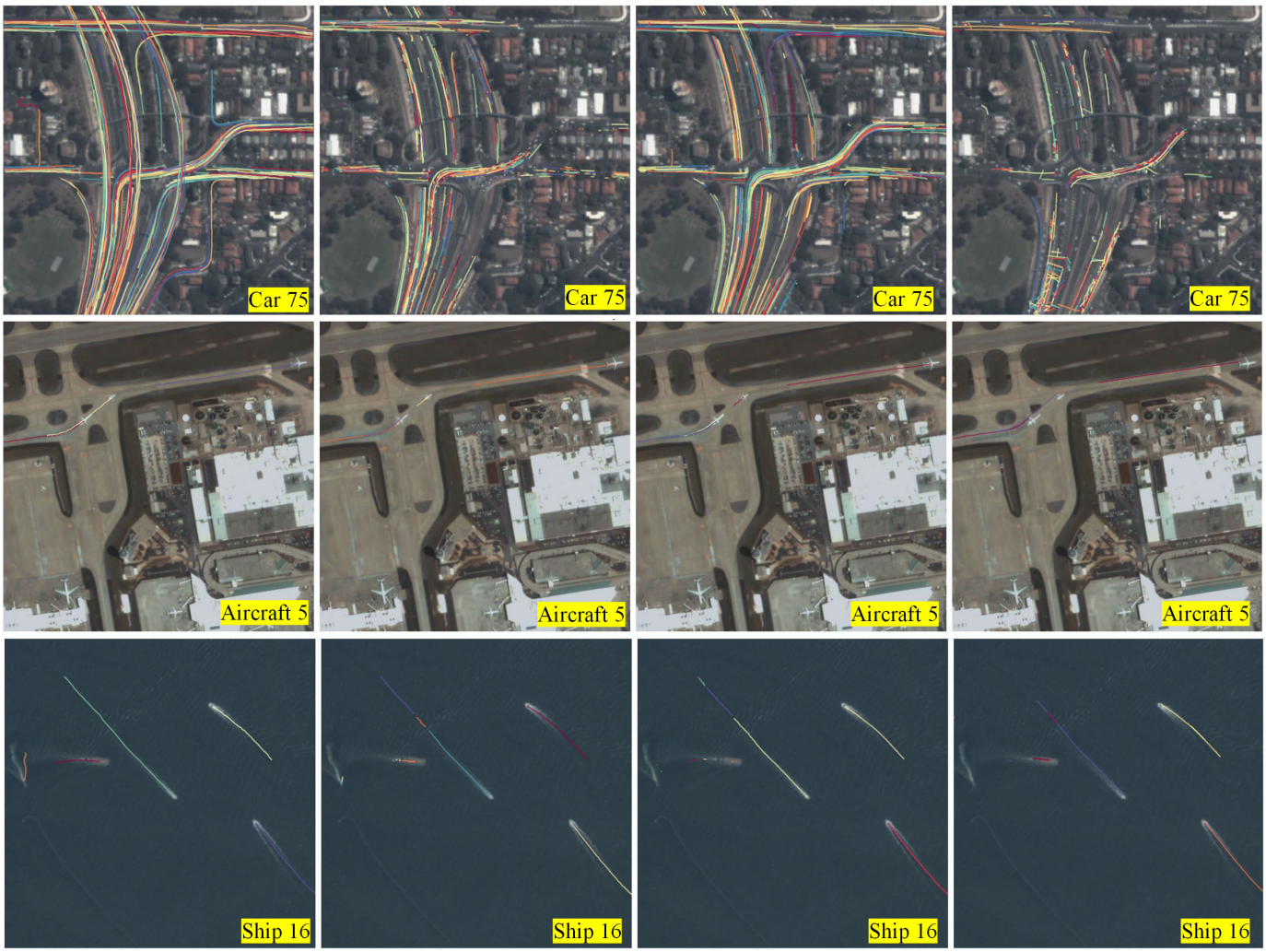


Fig. 11. Example of visualization results of object tracking trajectories. (a) GT. (b) CenterTrack. (c) DSFNet. (d) TGraM.

YOLACT. YOLACT uses a feature pyramid network to extract features at different scales from the input image, which are then processed by a series of convolutional layers to produce class-specific segmentation masks and bounding boxes for each detected object. One of the main advantages of YOLACT is its efficiency. It is able to achieve high accuracy while maintaining real-time performance, making it suitable for applications of video object segmentation. Considering the distribution characteristics of the center location and scale ratio of instances in the MS COCO dataset, Wang et al. [91] proposed a one-stage method called SOLO, which converts instance segmentation into semantic classification, and pixel-level mask classification. SOLOv2 [92] is an evolution of SOLO that utilizes a dynamic instance segmentation strategy to build unified high-resolution mask feature representations. In addition, SOLOv2 introduces matrix NMS, which utilizes matrix parallel operations to accelerate the NMS process.

b) Memory-based segmentation methods: Multimodal tracking Transformer (MTTR) [93] is an uncomplicated, memory-based approach for referring video object segmentation. MTTR formulates the task as a sequence prediction problem, without relying on text-related inductive bias modules. It accomplishes this by aligning video and text through a simple cross-entropy loss. VisTR [94] is a memory-based video instance segmentation framework that treats the video instance segmentation task as a simple end-to-end parallel sequence decoding or prediction problem. It employs an efficient instance sequence matching and segmentation strategy to supervise and segment instances at the whole sequence level. By adopting similarity learning as its core concept, VisTR simplifies the overall process and distinguishes itself from existing methods. EfficientVIS [95] is an end-to-end framework with a focus on efficient training and inference. Its central components are tracked queries and track suggestions that collaborate to segment RoIs in both space and time through iterative query video interactions. Compared to VisTR, EfficientVIS requires 15 times less training time while achieving state-of-the-art accuracy on YouTubeVIS benchmarks. Mask Transfiner [96] introduces a novel approach to instance segmentation by proposing a mask converter that utilizes quadtrees to represent image regions, rather than operating on a regular dense tensor. This method allows for the efficient

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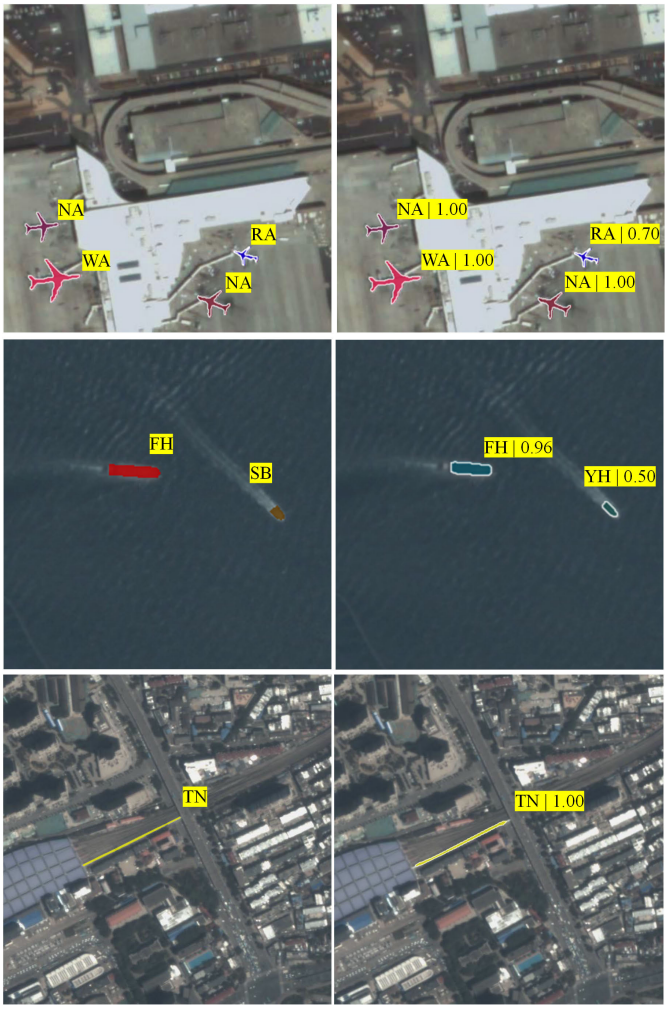


Fig. 12. Example of visualization results of object segmentation. (Left) Ground truth. (Right) HTC segmentation results.

processing of only error-prone tree nodes while self-correcting errors in parallel. The proposed approach outperforms existing instance segmentation methods on three popular benchmarks and significantly improves two-stage query-based frameworks. SeqFormer [97] is another method that aims to improve instance segmentation accuracy in videos. Unlike existing algorithms, SeqFormer creates a feature for each object in the video and introduces a query separation mechanism to extract object location information in each frame, which is then aggregated to represent each instance more efficiently at the video level. Idol [98] is another instance segmentation method that embeds instances and utilizes video history information to ensure algorithm stability, resulting in a performance level comparable to or higher than offline models.

2) Experiment Setup: First, we selected representative methods covering CNN-based segmentation methods, HTC [81], PointRend [86], SCNet [88], DetectoRS [87], Cascade Mask R-CNN [82], Mask R-CNN [79], Mask Scoring R- CNN [80], and QueryInst [89], YOLACT [90], and memory-based segmentation methods, MTTR [91], Mask Transfiner [96], IDOL [98], and SeqFormer [97], were trained, tested, and evaluated to compare the benchmark methods on the object segmentation task. All CNN-based methods were trained for 12 epochs with a batch size of 8. HTC, SCNet, and DetectoRS used a learning rate of $1e-2$ with a weight decay of $1e-4$. PointRend, Cascade Mask R-CNN, Mask R-CNN, and Mask Scoring R-CNN used a learning rate of $5e-3$ with a weight decay of $1e-4$. QueryInst used a learning rate of $5e-5$ with a weight decay of $1e-4$, while YOLACT had a learning rate of $1e-3$ with a weight decay of $5e-4$. All memory-based methods were trained with 20 epochs and a learning rate of $1e-3$ for the first 10 epochs and $5e-4$ for the last. All the methods utilized the default optimizer, and the performance was evaluated on the last epoch. The experiments were performed on a server with an Intel Xeon E5-2640 v4 CPU with 128-GB RAM and two Nvidia RTX GPUs.

3) Evaluation Metrics: For evaluation, we use AP and FPS metrics to quantitatively analyze the instance segmentation performance, with IoU set to 0.5 to be consistent with object detection.

4) Results and Analysis: The overall evaluation results of representative CNN- and memory-based object segmentation methods are shown in Table XII, and fine-grained category segmentation accuracy is shown in Table XIII. The visualization of the HTC segmentation results compared to the ground truth is shown in Fig. 12. Among them, the SOLO and SOLOv2 methods completely fail in the satellite video object segmentation task due to the large difference in object location and size distribution between the SAT-MTB dataset and the MS COCO dataset, so these two methods are not included in the results.

From the overall results, it is clear that the two-stage CNN-based approach results in higher accuracy and faster processing speed compared to the memory-based approach, while the one-stage CNN-based approach is close to the performance of the memory-based approach. Among them, HTC has the most outstanding performance in the experiment, and its horizontal object detection is also close to the best SCNet and DetectoRS, but the FPS is relatively low, which improves the accuracy but loses the model inference speed.

The reasons for the better performance of two-stage CNN-based approaches in the satellite video object segmentation task are given as follows. 1) two-stage CNN-based approaches adopt a top-down detection-segmentation model, which reduces the full-image search time and incorrect segmentation regions; 2) the satellite video object features are weak, and memory-based approaches extract deep features that easily lose shallow information, such as object shape and location; and 3) memory-based methods introduce interframe information transfer, which introduces redundant information for still object segmentation, resulting in lower accuracy and slower inference. Using the mAP metric from the first-level classification, aircraft positioning is relatively accurate, with high mask segmentation accuracy and poor fine-grained classification. Ship positioning accuracy is low, but fine-grained classification is also poor. The positioning accuracy and mask segmentation accuracy of the train are both low. In the fine-grained segmentation results, categories with high segmentation accuracy are WA, NA, RA, FA, and NV. These categories of objects are relatively abundant in both the training and test

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TABLE XII
COMPARISON OF OVERALL EVALUATION RESULTS FOR OBJECT SEGMENTATION

	AP(seg)	AP(bbox)	FPS	AP		
				Aircraft	Ship	Train
HTC [81]	0.347	0.344	7.8	0.765	0.270	0.086
PointRend [86]	0.278	0.293	10.0	0.760	0.233	0.266
SCNet [88]	0.312	0.375	6.8	0.808	0.277	0.015
DetectoRS [87]	0.303	0.355	3.4	0.762	0.262	0.057
Cascade Mask R-CNN [84]	0.293	0.333	10.0	0.766	0.281	0.008
Mask R-CNN [79]	0.266	0.305	13.0	0.763	0.280	0.040
Mask Scoring R-CNN [80]	0.233	0.269	13.0	0.822	0.238	0.040
QueryInst [89]	0.129	0.145	7.6	0.290	0.125	0.000
YOLACT [90]	0.136	0.197	16.0	0.250	0.206	0.000
MTTR [94]	0.194	-	1.3	0.410	0.136	0.076
Mask Transfiner [96]	0.176	-	2.1	0.337	0.165	0.046
IDOL [98]	0.105	-	0.5	0.259	0.089	0.013
SeqFormer [97]	0.103	-	0.7	0.294	0.061	0.006

TABLE XIII
COMPARISON OF FINE-GRAINED BENCHMARK EVALUATION RESULTS FOR OBJECT SEGMENTATION

	WA	NA	RA	FA	CA	SB	YH	CS	FH	NV	OS	TN
HTC [81]	0.710	0.550	0.737	0.478	0.303	0.001	0.357	0.000	0.093	0.633	0.215	0.086
PointRend [86]	0.641	0.553	0.616	0.378	0.234	0.000	0.318	0.000	0.098	0.027	0.204	0.266
SCNet [88]	0.745	0.663	0.893	0.551	0.288	0.001	0.387	0.000	0.090	0.003	0.110	0.015
DetectoRS [87]	0.763	0.579	0.771	0.558	0.248	0.005	0.356	0.000	0.165	0.001	0.135	0.057
Cascade Mask R-CNN [84]	0.601	0.530	0.700	0.374	0.240	0.002	0.388	0.000	0.043	0.306	0.326	0.008
Mask R-CNN [79]	0.669	0.527	0.605	0.408	0.248	0.000	0.369	0.000	0.048	0.155	0.141	0.016
Mask Scoring R-CNN [80]	0.621	0.455	0.690	0.349	0.165	0.000	0.331	0.000	0.000	0.021	0.118	0.040
QueryInst [89]	0.163	0.225	0.542	0.113	0.105	0.014	0.145	0.000	0.005	0.003	0.231	0.000
YOLACT [90]	0.426	0.113	0.195	0.374	0.049	0.006	0.264	0.000	0.000	0.000	0.210	0.000
MTTR [94]	0.231	0.318	0.531	0.265	0.094	0.002	0.216	0.000	0.012	0.036	0.198	0.076
Mask Transfiner [96]	0.416	0.248	0.485	0.134	0.076	0.000	0.135	0.000	0.009	0.016	0.154	0.046
IDOL [98]	0.130	0.162	0.246	0.165	0.002	0.000	0.102	0.000	0.000	0.028	0.126	0.013
SeqFormer [97]	0.162	0.127	0.298	0.211	0.016	0.000	0.006	0.000	0.000	0.038	0.209	0.006

sets of the dataset and have relatively obvious texture features. The SB and CS have poor segmentation accuracy due to their similar object appearance but are prone to misclassification when they lose their scale reference. Moreover, the color of CSs is close to the sea surface, and some CSs are misclassified as background. Trains, on the other hand, are more difficult to segment because of their particular serpentine shape, which can be rigidly deformed. In two-stage segmentation methods, it is difficult to locate objects in the first stage. Especially for trains where the two ends are far from the center point and the two sides are close to the center point, the accuracy of segmentation is lower when segmentation is performed after positioning. The performance of these baseline models still has a lot of room for improvement. The fine-grained classification of aircraft and ships needs to be improved, and the localization accuracy of ships and trains needs to be further improved, while the segmentation accuracy of the train is a major challenge. In addition, the robustness of the segmentation method should be further optimized. In addition, the dataset characteristics and experimental results show that the dataset not only includes different object types and different scenario scales but also can be used to solve the problems of object occlusion, object rotation, object deformation, illumination changes, object loss, and so on. Compared with the general video object segmentation, this dataset provides data sources with different perspectives, which can better cope with the problems of small objects and weak object texture. Compared with the satellite video single object segmentation dataset, this dataset provides more and more comprehensive annotation labels, which makes up for the shortcomings of the existing video object segmentation dataset and provides a sufficient database for the research of video object segmentation.

V. CONCLUSION

In this article, we build SAT-MTB, the first large-scale satellite video dataset that combines object detection, object tracking, and object segmentation, providing rich experimental material and data benchmarks for research in various tasks of satellite video. The SAT-MTB is the largest satellite video dataset in terms of the number of videos, the types of tasks supported, and the categories of objects, which can support high-quality and rich annotated samples for the training and validation of various algorithms. In addition, the dataset provides more difficult subtasks, such as oriented objects and fine

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graining, maximizing the number of applications for satellite video research.

In addition, this article also evaluates and analyzes the performance of representative methods of object detection, object tracking, and object segmentation on SAT-MTB. The experimental results show that the proposed dataset is extremely challenging in detecting objects with fine-grained categories and extreme aspect ratios, tracking objects with dense distribution of multicategory and similar appearance, and segmenting objects with small size and weak texture, where existing representative methods perform poorly or even fail and need to be combined with the data characteristics of satellite video for algorithm improvement and optimization. The public benchmark of multitask algorithms for satellite video object detection, object tracking, and object segmentation established in this article will significantly advance research in intelligent processing and analysis of satellite video and related applications.

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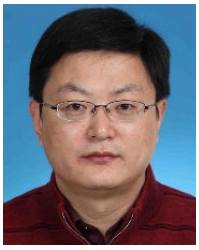
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